Zeina Helali - 202302006

CSAI 422 – Assignment 4
Retrieval-Augmented Generation with Advanced Techniques

1. Introduction

Retrieval-Augmented Generation (RAG) is a method that enhances the robustness of a language model by applying the process of information retrieval along with text generation. In contrast to a standard approach when only knowledge extracted by a language model is taken into account, RAG systems perform document retrieval from an external corpus to provide context for answering a query.

The task assigned in this case is to design an end-to-end RAG pipeline that will work based on PopQA dataset for factual question answering. The process will begin with the development of a basic dense retriever that will be improved through the application of such retrieval techniques as query expansion, hybrid retrieval, reranking, citation-grounded answer generation, and self-reflection.

The primary purpose of this assignment is to investigate the role of retrieval quality and ranking in generating answers and additional techniques that may increase system reliability.

2. Dataset and Corpus

The dataset used in this assignment is PopQA, a benchmark dataset designed for factual question answering tasks. The dataset contains natural language questions together with their possible answers and metadata about the entities involved. The dataset was loaded from HuggingFace and inspected to understand its structure. The important fields include:

- question – the natural language question
- possible_answers – list of correct answers
- s_wiki_title / o_wiki_title – Wikipedia page titles related to the subject and object of the question

Because the full dataset is large, a subset of 50 questions was selected to make the experiments computationally manageable.

To build the retrieval corpus, Wikipedia pages were collected using the subject and object titles from the dataset. Each Wikipedia page was downloaded and divided into smaller text passages (chunks) so that they could be efficiently indexed and retrieved later. Each passage was stored with a unique identifier (passage_id) and its associated metadata such as the page title and text

content. Keeping these identifiers is important because the final generated answers must cite the passages used as evidence.

```
Generating test split: 100% 14267/14267 [00:00<00:00, 49894.41 examples/s]
Dataset shape: (14267, 17)
Columns:
['id', 'subj', 'prop', 'obj', 'subj_id', 'prop_id', 'obj_id', 's_aliases', 'o_aliases', 's_uri', 'o_uri', 's_wiki_title', 'o_wiki_title', 's
```

	id	subj	prop	obj	subj_id	prop_id	obj_id	s_aliases	o_aliases	s_uri
0	4222362	George Rankin	occupation	politician	1850297	22	2834605	["George James Rankin"]	["political leader","political figure","polit....	http://www.wikidata.org/entity/Q5543720
1	4725190	John Mayne	occupation	journalist	2079053	22	663400	[]	["journo","journalists"]	http://www.wikidata.org/entity/Q6247345
2	4382392	Henry Feilden	occupation	politician	1925450	22	2834605	["Henry Master Feilden"]	["political leader","political figure","polit....	http://www.wikidata.org/entity/Q5725578
3	4822110	Kathy Saltzman	occupation	politician	2122743	22	2834605	[]	["political leader","political figure","polit....	http://www.wikidata.org/entity/Q6377295
4	4011112	Eleanor Davis	occupation	cartoonist	1752619	22	68412	["Eleanor McCutcheon Davis"]	["graphic artist","animator","illustrator"]	http://www.wikidata.org/entity/Q5354261

```
s_pop', 'o_pop', 'question', 'possible_answers']
```

	o_uri	s_wiki_title	o_wiki_title	s_pop	o_pop	question	possible_answers
	http://www.wikidata.org/entity/Q82955	George Rankin	Politician	142	25692	What is George Rankin's occupation?	["politician", "political leader", "political ...
	http://www.wikidata.org/entity/Q1930187	John Mayne	Journalist	236	24952	What is John Mayne's occupation?	["journalist", "journo", "journalists"]
	http://www.wikidata.org/entity/Q82955	Henry Feilden (Conservative politician)	Politician	58	25692	What is Henry Feilden's occupation?	["politician", "political leader", "political ...
	http://www.wikidata.org/entity/Q82955	Kathy Saltzman	Politician	127	25692	What is Kathy Saltzman's occupation?	["politician", "political leader", "political ...
	http://www.wikidata.org/entity/Q1114448	Eleanor Davis	Cartoonist	317	9649	What is Eleanor Davis's occupation?	["cartoonist", "graphic artist", "animator", "...

Figure 1: Structure of the PopQA dataset showing the question and possible answer fields.

3. Baseline Dense Retrieval

The first step in the pipeline is to build a baseline dense retrieval system. Dense retrieval uses vector embeddings to represent text passages and queries in a shared semantic space.

The embedding model used in this project is: sentence-transformers/all-MiniLM-L6-v2

This model was selected because:

- It produces high-quality semantic embeddings
- It is lightweight and fast
- It works well for retrieval tasks

All passages in the corpus were embedded using this model and stored in a FAISS vector index, which allows efficient similarity search.When a question is given, the system embeds the query and retrieves the top-k most similar passages from the index using cosine similarity.

```
Embedding model: sentence-transformers/all-MiniLM-L6-v2
Embedding dimension: 384
Total indexed passages: 3228
```

Figure 2: Dense retrieval index statistics after embedding the corpus passages.

```
QUESTION: Who was the screenwriter for On the Town?
Montgomery_Tully_1 | 1967. He also worked in television, directing episodes of shows including Edgar Wallace Mysteries, Kraft Mystery Theatre
The_City_(1916_film)_0 | The City is a lost 1916 silent film based on Clyde Fitch's 1909 play, The City. It was distributed by the World Film
Last_Night_(2010_film)_9 | that their struggle was believable". After a negative response from a Warner Bros. studio executive on the script'

QUESTION: What is the religion of Juan Soldevilla y Romero?
Juan_Soldevilla_y_Romero_0 | Juan Soldevila y Romero (29 October 1843 – 4 June 1923) was a Spanish Cardinal of the Roman Catholic Church who
Juan_Soldevilla_y_Romero_1 | canon law at the seminary in Tuy. Soldevila served as a curate in three parishes in the Archdiocese of Valladoli
Juan_Soldevilla_y_Romero_4 | of anarchist leader Salvador Seguí. He is buried at the Basilica of Our Lady of the Pillar. == References == ==

QUESTION: What genre is Haddaway?
The_Album_(Haddaway_album)_0 | The Album (also known as Haddaway in North America and L'Album in France) is the debut album by Trinidadian-Ge
The_Album_(Haddaway_album)_3 | on The Album, additional backing vocals on The Album 2nd Edition by Elkie Schlimbach, Rena T. Otta Alex Trime
Washington,_D.C._78 | was one of the most crucial independent labels in the genesis of 1980s punk and eventually indie rock in the 1990s. Mod
```

Figure 3: Example results from the baseline dense retriever.

4. Baseline Retrieval Evaluation

To measure the performance of the retriever, several retrieval metrics were used.

- Recall@k: Recall@k measures whether the correct answer appears in the top-k retrieved passages.
- Precision@k: Precision@k measures the proportion of retrieved passages that contain the correct answer.
- Mean Reciprocal Rank (MRR): MRR measures how early the first correct passage appears in the ranking.

These metrics help evaluate how well the retriever can locate relevant passages before generating answers.

```
100%|██████████| 50/50 [00:01<00:00, 31.72it/s]
```

	Recall@1	Recall@3	Recall@5	Precision@1	Precision@3	Precision@5	MRR
Dense Baseline	0.62	0.74	0.82	0.62	0.346667	0.32	0.699

Figure 4: Retrieval performance of the baseline dense retriever.

5. Query Expansion

Query expansion is used to improve retrieval by adding additional keywords or related terms to the original query. In this assignment, I implemented a simple expansion strategy by adding common factual search terms such as:

- biography
- history
- known for
- is
- was

The goal of this technique is to provide more context to the retrieval model and increase the chance of matching relevant passages.

Query expansion improved retrieval in some cases, especially when the original question was short or ambiguous. However, it also introduced noise when the added keywords caused unrelated passages to appear in the results.

```
Original: Who was the screenwriter for On the Town?
Expanded: Who was the screenwriter for On the Town? biography history known for was is

Original: What is the religion of Juan Soldevilla y Romero?
Expanded: What is the religion of Juan Soldevilla y Romero? biography history known for was is

Original: What genre is Haddaway?
Expanded: What genre is Haddaway? biography history known for was is

Original: Who was the composer of Chances Are?
Expanded: Who was the composer of Chances Are? biography history known for was is

Original: Who was the producer of The Pioneers?
Expanded: Who was the producer of The Pioneers? biography history known for was is
```

Figure 5: Examples comparing original queries with expanded queries.

6. Hybrid Retrieval

Hybrid retrieval combines dense retrieval and lexical retrieval. In this system:

Dense retrieval: semantic similarity
BM25: keyword matching

Both retrieval results were combined using Reciprocal Rank Fusion (RRF), which merges ranked lists from different retrievers. This approach improves performance because dense models capture semantic similarity while lexical models capture exact keyword matches.

```
100%|██████████| 50/50 [00:02<00:00, 24.90it/s]
```

	Recall@1	Recall@3	Recall@5	Precision@1	Precision@3	Precision@5	MRR
Dense Baseline	0.62	0.74	0.82	0.62	0.346667	0.320	0.699000
Dense + Query Expansion	0.52	0.70	0.76	0.52	0.340000	0.304	0.619667
Hybrid BM25 + Dense	0.56	0.76	0.82	0.56	0.406667	0.340	0.666333

Figure 6: Comparison of dense retrieval, query expansion, and hybrid retrieval.

7. Reranking

Although retrieval methods return relevant passages, their ranking may still be imperfect. To improve ranking quality, a cross-encoder reranker was used: cross-encoder/ms-marco-MiniLM-L-6-v2

The reranker evaluates each query–passage pair and assigns a more accurate relevance score.The passages are then sorted again using these scores.This step significantly improves the quality of the top retrieved passages.

```
QUESTION: Who was the screenwriter for On the Town?

Before reranking:
[P1] On_the_Town_(film)_0 | On the Town is a 1949 American Technicolor musical film with music by Leonard Bernstein and Roger Edens and book
[P2] Montgomery_Tully_1 | 1967. He also worked in television, directing episodes of shows including Edgar Wallace Mysteries, Kraft Mystery Th
[P3] Bury,_Greater_Manchester_30 | Metropolitan Police, who was born in Bury and was responsible for the repeal of the Corn Laws in 1846. Hun
[P4] The_City_(1916_film)_0 | The City is a lost 1916 silent film based on Clyde Fitch's 1909 play, The City. It was distributed by the World
[P5] Bury,_Greater_Manchester_0 | Bury (, ) is a market town on the River Irwell in the Metropolitan Borough of Bury, Greater Manchester, Eng

After reranking:
[P1] On_the_Town_(film)_0 | On the Town is a 1949 American Technicolor musical film with music by Leonard Bernstein and Roger Edens and book
[P2] Betty_Comden_3 | wrote the book and lyrics, which included sizable parts for themselves (as "Claire" and "Ozzie"). Their next musical, B
[P3] Musical_film_3 | regularly premiered. These works included: Meet Me in St. Louis (1944), Easter Parade (1948), On the Town (1949), An Am
[P4] Last_Night_(2010_film)_9 | that their struggle was believable". After a negative response from a Warner Bros. studio executive on the sc
[P5] The_Night_They_Raided_Minsky's_10 | that the novel would be adapted by screenwriter Edward Chodorov. Later in the year, however, Key had
```

Figure 7: Comparison of retrieved passages before and after reranking.

8. Citation-Grounded Answer Generation

After retrieval, the best passages are used as context for generating answers, the system uses a prompt that forces the model to:

- use only retrieved passages
- include citations such as [P1], [P2]
- avoid unsupported claims
- say when evidence is insufficient

And each answer is generated based only on the retrieved passages and includes explicit references to them.

```
================================================================================
QUESTION: Who was the screenwriter for On the Town?

RETRIEVED PASSAGES:
[P1] On_the_Town_(film)_0 | On the Town is a 1949 American Technicolor musical film with music by Leonard Bernstein and Roger Edens and bool
[P2] Betty_Comden_3 | wrote the book and lyrics, which included sizable parts for themselves (as "Claire" and "Ozzie"). Their next musical,
[P3] Musical_film_3 | regularly premiered. These works included: Meet Me in St. Louis (1944), Easter Parade (1948), On the Town (1949), An
[P4] Last_Night_(2010_film)_9 | that their struggle was believable". After a negative response from a Warner Bros. studio executive on the
[P5] The_Night_They_Raided_Minsky's_10 | that the novel would be adapted by screenwriter Edward Chodorov. Later in the year, however, Key h

ANSWER:
Based on the retrieved evidence, the answer is most likely supported by [P1]. On the Town is a 1949 American Technicolor musical film with
================================================================================
QUESTION: What is the religion of Juan Soldevilla y Romero?

RETRIEVED PASSAGES:
[P1] Juan_Soldevilla_y_Romero_0 | Juan Soldevila y Romero (29 October 1843 – 4 June 1923) was a Spanish Cardinal of the Roman Catholic Chur
[P2] Juan_Soldevilla_y_Romero_1 | canon law at the seminary in Tuy. Soldevila served as a curate in three parishes in the Archdiocese of Va
[P3] Juan_Soldevilla_y_Romero_2 | for the assistance of victims of a cholera epidemic. On 14 February 1889 Soldevila was appointed Bishop o
[P4] Juan_Soldevilla_y_Romero_3 | del Popolo in the consistory of 15 December 1919. He received his cardinal's biretta from King Alphonse X
[P5] Juan_Soldevilla_y_Romero_4 | of anarchist leader Salvador Seguí. He is buried at the Basilica of Our Lady of the Pillar. == References

ANSWER:
Based on the retrieved evidence, the answer is most likely supported by [P1]. Juan Soldevila y Romero (29 October 1843 – 4 June 1923) was a
--------------------------------------------------------------------------------
```

Figure 8: Example grounded question answering results with passage citations.

9. Error Analysis

An error analysis was performed on several cases where the system failed or produced incomplete answers. The main types of errors observed were:

1. **Retrieval failures**
   The correct passage was not retrieved.
2. **Ranking errors**
   The correct passage was retrieved but ranked too low.
3. **Generation issues**
   The model produced incomplete answers despite correct evidence.
4. **Corpus limitations**
   Some answers were not clearly present in the collected Wikipedia passages.

Improving the retrieval corpus or using stronger reranking models could reduce these errors.

	Question	Gold Answers	Retrieval Hit	Likely Failure Source	Possible Fix
0	Who was the screenwriter for On the Town?	[betty comden, basya cohen, adolph green]	True	Generation or prompting issue if answer is sti...	Improve prompt instructions or use stronger LLM.
1	What is the religion of Juan Soldevilla y Romero?	[catholic church, roman catholic church, churc...	True	Generation or prompting issue if answer is sti...	Improve prompt instructions or use stronger LLM.
2	What genre is Haddaway?	[eurodance, eurodance, euro dance]	True	Generation or prompting issue if answer is sti...	Improve prompt instructions or use stronger LLM.
3	Who was the composer of Chances Are?	[maurice jarre, mauricealexis jarre]	True	Generation or prompting issue if answer is sti...	Improve prompt instructions or use stronger LLM.
4	Who was the producer of The Pioneers?	[franklyn barrett, walter franklyn barrett]	True	Generation or prompting issue if answer is sti...	Improve prompt instructions or use stronger LLM.

Figure 9: Failure analysis examples and possible causes.

10. Self-Reflective RAG

A self-reflection stage I added after answer generation. If problems are detected, the system revises the answer using the same retrieved passages.This step helps improve reliability and reduces unsupported claims.

This stage evaluates whether:

- the answer is grounded in evidence
- the citations are correct
- the answer is complete

```
QUESTION:
Who was the screenwriter for On the Town?

ORIGINAL ANSWER:
Based on the retrieved evidence, the answer is most likely supported by [P1]. On the Town is a 1949 American Technicolor musical film with mu

REFLECTION OUTPUT:
Critique: The answer should clearly cite the retrieved passages and avoid unsupported claims.
Revised Answer: The answer should be based mainly on [P1]. On the Town is a 1949 American Technicolor musical film with music by Leonard Bern
```

Figure 10: Example showing answer improvement after self-reflection.

11. Comparative Evaluation

The final system compared four configurations and each configuration was evaluated using retrieval metrics:

1. Baseline dense retrieval
2. Dense retrieval + query expansion
3. Hybrid retrieval + reranking
4. Final RAG system with self-reflection

	Recall@1	Recall@3	Recall@5	Precision@1	Precision@3	Precision@5	MRR
Baseline Dense Retrieval	0.62	0.74	0.82	0.62	0.346667	0.320	0.699000
Dense Retrieval + Query Expansion	0.52	0.70	0.76	0.52	0.340000	0.304	0.619667
Hybrid Retrieval	0.56	0.76	0.82	0.56	0.406667	0.340	0.666333
Hybrid Retrieval + Reranking	0.84	0.86	0.86	0.84	0.500000	0.408	0.850000
Final RAG + Self-Reflection	0.84	0.86	0.86	0.84	0.500000	0.408	0.850000

Figure 11: Final comparison of retrieval performance across all systems.

12. Conclusion

This assignment showed how the use of various retrieval and ranking methods can impact the effectiveness of a RAG model.

The dense retriever served as a good baseline, but it failed to retrieve some of the relevant documents. Query expansion was useful in some cases but at times led to the introduction of noise. Hybrid retrieval method combined lexical and semantic retrieval methods and therefore led to better results.

Use of reranking improved the quality of the top retrieved documents. Lastly, the self-reflection stage helped enhance answer reliability through citation verification. Generally, hybrid retrieval models with reranking and self-reflection performed best since it generated the most reliable answers. Future improvements can include working with a bigger corpus, enhancing the reranking technique, and query expansion methods.